BNUUY'S SHIP

Brian Moniaga, Chinmay Bhansali,
Clare Pan, Dayshaun Lee, Lily Zhang

1

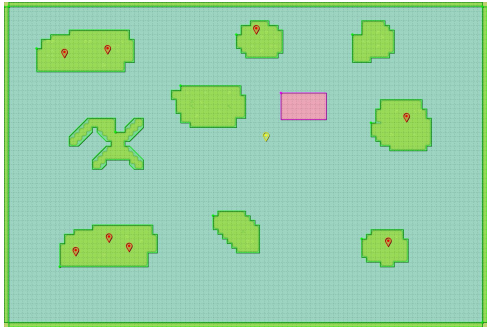
Required Elements

Gameplay

Keyboard/mouse control

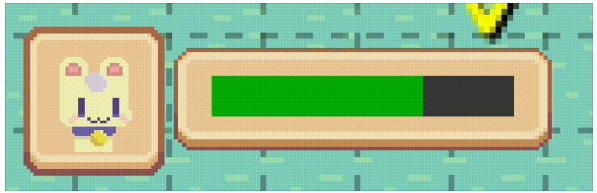
- In the menu, left click ‘Play’ button to start game
- In player mode (game starts in this mode)
    - Press WASD to control player movement

Rendering



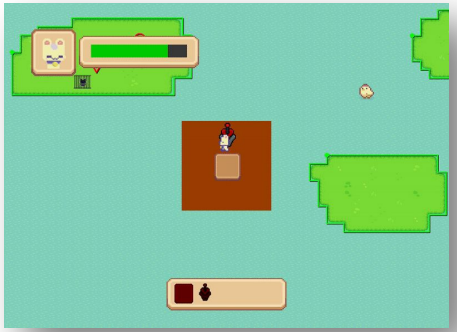
Textured geometry

- Load and render map level
- Animated player sprites



Basic 2D transformation

- Entity motion's position transformation
- Scale textures to screen



Key-frame/state interpolation

- Smooth interpolation of animated player, enemy movement

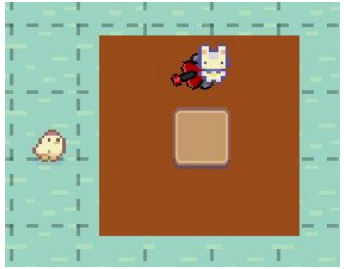
Gameplay

Keyboard/mouse control

To control the ship:

- Move Bnuuy to the steering wheel (the plate in the middle of the ship)
- Press space to toggle use station
- Press WASD to control ship

To control weapon station



- Move Bnuuy to the cannon
- Press space to toggle use station
- Aim using mouse and left click to shoot

Gameplay

Keyboard/mouse control

To rearrange the station modules on the ship:

- To remove station from ship: right click the station on ship to put into inventory
- To put station back to the ship: left click the item in the inventory to select and left click on the position to place on ship

Gameplay

Well-defined game-space boundaries

- Player can't walk off the ship
- Ship can't move onto islands

Simple collision detection & resolution

- Spherical collision
- Polygon collision

Random/coded action

- Island and enemy spawn location are initialized inside map generation

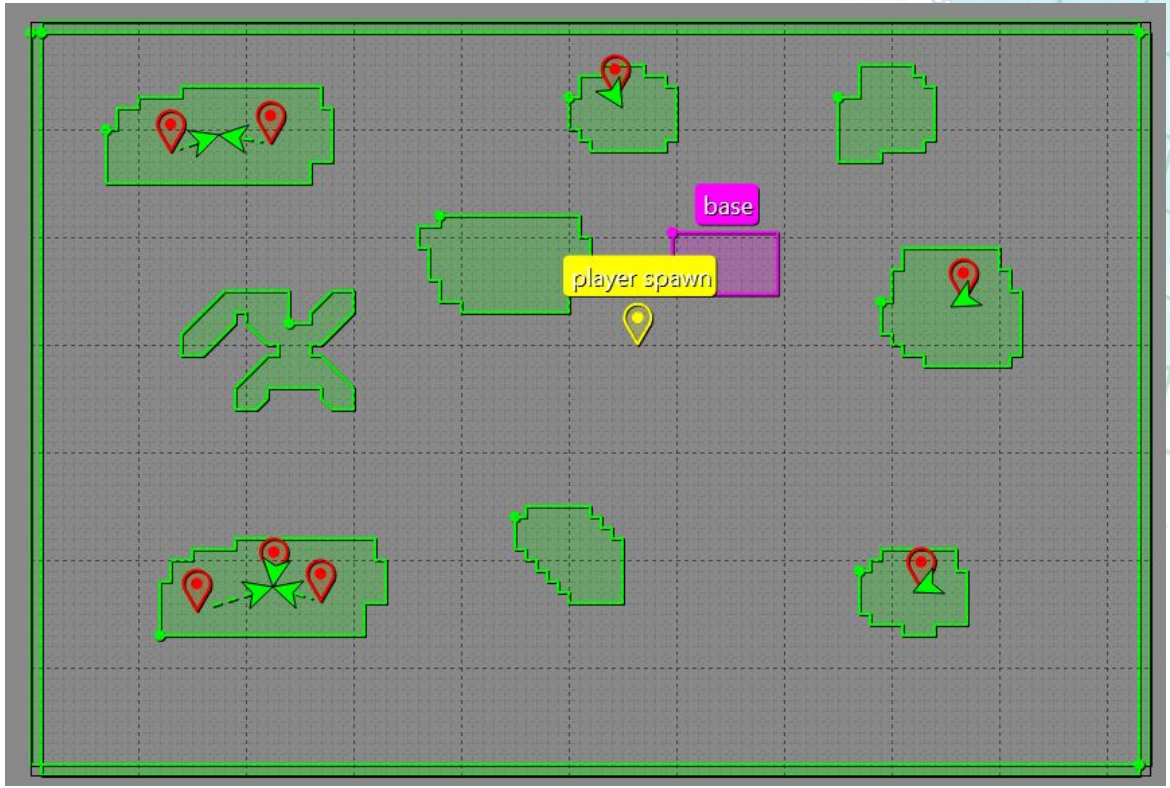
2

Creative Elements

External integration - Map Objects

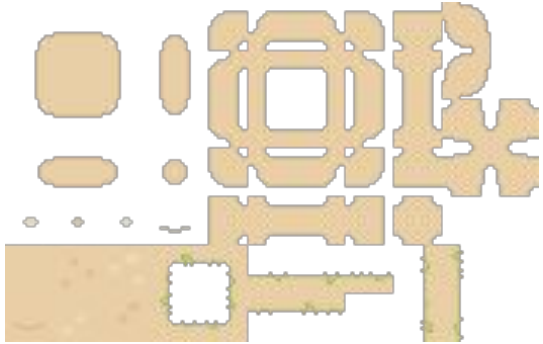
Tileson (A modern cross-platform json-parser for parsing Tiled maps)

- Loading objects from map:
  - Player spawn point
  - Enemy spawn points
  - Island/base positions and polygon vertices



Textured Map

- Composited into single texture with Tiled
- Grass for islands
- Dirt for base
- Repeating water texture for background



Asset: https://cupnooble.itch.io/sprout-lands-asset-pack

Bnuuy Sprites

Asset: https://cupnooble.itch.io/sprout-lands-ui-pack

Precise Collision

Advanced polygon-polygon collision

- Detects collision based on island polygons
- Uses location of previous frame to ensure consistency

Completed tasks

1

Brian

- Ship module system
- Player Movement
- UI system
- Scene Manager
- Island Collision resolution

2

Chinmay

- Island, Ship and Player Collision Detection and resolution

3

Clare

- Enemy rendering and movement
- Bunny saving mechanics
- Projectile collision

4

Dayshaun

-Map loading
-Map rendering
-Collision detection
-World boundary

5

Lily

- Physics based ship movement
- Ship rendering
- Camera movement

Thanks!

Credits: SlidesCarnival for the presentation template